Distributed Power Management for 22nm AI Processor with Event-driven Exponential Dual-loop LDOs and Online Sparsity-Aware Droop Mitigation

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Abstract

A distributed power management solution is presented for a 22nm digital compute-in-memory (DCIM) based AI processor featuring distributed dual-loop digital LDOs with event-driven exponential control for fast response, online learning-based sparsity-aware droop mitigation, and workload-driven current balance scheme among LDOs. The work demos the promising opportunity of deploying distributed management for robust and efficient AI computing, where the above techniques lead to >71.6% droop reduction for diverse AI workloads and up to 33.3% energy saving or 22.5% performance improvement.

Introduction

AI processors or accelerators are essential for AI computing, while they consume tens of Watts of power [1] and require delicate power management. Most prior AI processors only focused on architecture-level innovations and simply used board-level [2] power monitoring or regulation. Distributed power management is a promising approach for large-scale design [3-8], while several unresolved challenges remain for AI processors (Fig. 1). First, the massive parallel MAC operations in AI processors generate amps-level transient current, which incurs significant voltage droop (>150mV). Second, the large model sparsity (10% to >90%) directly impacts digital toggle rates and further exacerbates the pessimistic voltage guardband. Third, the current dissipation is imbalanced across LDOs during runtime, leading to a reduced effective number and power delivery efficiency.

In this work, we present a distributed power management for AI processors with three key features: 1) distributed dual-loop digital LDOs (DLDOs) using event-driven exponential (EE) control for fast response, 2) online learning-based sparsity-aware droop mitigation to optimize voltage regulation, 3) a workload-driven current balance scheme to regulate the current sharing imbalance among DLDOs.

Distributed Power Management for AI Processor

Fig. 2 provides an overview of AI processors with distributed power management. The AI inference is performed in a 4TOPS NPU design, which consists of 3×3 DCIM clusters. The chip also includes a host RISC-V CPU, global memory, and a voltage recorder for voltage sensing. The distributed LDOs feature a dual loop control: a local-DLDO (L-LDO) loop within each DCIM cluster and a global-DLDO (G-LDO) loop centrally monitors all DLDOs and takes over control when no events are triggered. An online learning sparsity-aware unit is adopted to collect runtime information and optimize droop using a learning algorithm. A current balance module is designed using local mismatch cancellation and global workload-driven power switch balance to adjust each L-LDO.

Fig. 3 illustrates the details of distributed event-driven dual-loop DLDOs. The G-LDO global loop is used as slow mode control, including a main comparator and a four-state integrator. The L-LDO local loop is designed for fast mode control, including two INV-based comparators (consume less static power than 2.6-bit ADC in [4]) and an event-driven exponential controller (EEC). In steady-state, the DLDOs operate in slow mode and are controlled by the sum of a static C_{LOCAL} and C_{GLOBAL}, which is tuned by integral factor K_I and adds or subtracts half of accumulation code (Cnt) in the global four-state integrator for delay compensation. The EEC is triggered once the INV-based comparators detect $V_{OUT}<V_{REFL}$ (Fig. 4). The event signal has a critical path latency of less than 400ps, achieving faster response than prior clock-based exponential LDO [10] and is well-suited for large transient current regulation for AI computing. Once an event is detected, the power switches are tied high (C_{TieHi}) (t_1), and C_{LOCAL} is increased by exponential factor K_E. After V_{OUT} rises above V_{REFL}(t_2), the updated C_{LOCAL} regulates local DLDOs. Before I_{DLDO} reaches I_{LOAD}, EEC keeps increasing C_{LOCAL} by twice as large as $K_E\times2^{n-1}$ (n-th control code changes). If V_{OUT} raises over V_{REFH} (t_3), an overshoot event is detected, and C_{LOCAL} is reduced until DLDOs return to steady-state (t_4).

Fig. 5 illustrates our learning-based sparsity-aware droop mitigation. Voltage droop/overshoot are detected and used as learning rewards, e.g. larger droop leads to more negative rewards in the reward calculator. Runtime weight sparsity and input toggle are monitored by toggle detectors [11] to index a learning table, which stores learned rewards V_{DRP}/V_{OS} and configuration actions. These actions set the reference voltages V_{REFH} and V_{REFL} for L-LDOs, V_{OS} and V_{DRP} for the droop detectors, and codes C_{TieLo} and C_{TieHi} for EEC. The online learning involves two steps: droop mitigation (S.1) and droop evaluation (S.2) (Fig. 6). S.1 gradually narrows the event detection window between V_{REFH} and V_{REFL} to reduce peak-to-peak V_{OUT} by triggering EEC with a shorter response time. S.2 evaluates the droop magnitude by adjusting V_{OS} and V_{DRP}. The optimal settings can be learned for different sparsity during runtime to minimize droop while maintaining LDO stability. The learned settings can be used in DVFS to adjust in real-time based on the workload for better efficiency/performance.

Fig. 7 shows workload-driven current balance and mismatch cancellation. In our work, the comparator mismatch among L-LDOs is canceled by periodic calibration, i.e. switching coupling capacitor C_{OS} between $V_{REFH/L}$ and V_{OUT} to maintain identical references. To balance the current for different workloads, a power switch balance module collects C_{LOCAL} from all L-LDOs and generates the current balance adjustment code C_{CB} for each L-LDO in slow mode. The CPU monitors runtime workload sparsity and adjusts the power switch balance strategy to match workload distributions. This strategy reduces current imbalance σ among L-LDOs and maintains a constant total C_{LOCAL}.

Measurement Results

We present measurement results on our 22nm AI processor running a Transformer block from SDv1.5 model (Fig.8). With the fast-response EEC local loop, a 44.3% droop reduction is obtained compared to conventional control loop under dense workloads. The L-LDO has a fast transient response of 390mA/200ps for 35mV undershoot and 170mA/200ps for 15.7mV overshoot, with 99.9% peak current efficiency. Fig. 9 shows that by enabling our online learning feature, 49-71.3% voltage droop is mitigated in 30 learning iterations for different sparsity. Fig. 10 shows the die photo and overall benefits. Our approach achieves 12.3-22.5% performance improvement or 21.1-33.3% energy saving with a reduced voltage guardband under different sparsity. Dense AI workloads benefit more from larger droop mitigation, while sparse workloads gain the majority of the improvement by our learning-based regulation.

Table 1 presents comparisons with related designs. Compared to prior power management for CPU/SoC [12,13,9], our approach first demonstrates a distributed power management solution for AI computing with the largest droop mitigation benefits. More than 71.6% of droop reduction is obtained without performance loss and over 2× higher than prior work [12]. Our approach is also orthogonal to prior AI SoC [2, 9] and can be combined to improve further performance or efficiency. Compared to distributed LDOs with synthetic resistor loads, our EEC LDOs achieve a >10× FoM boost for 0.2ns T_{EDGE} [4]. Additionally, we adjust the current balancing based on workload, improving the local power supply compared to [3].

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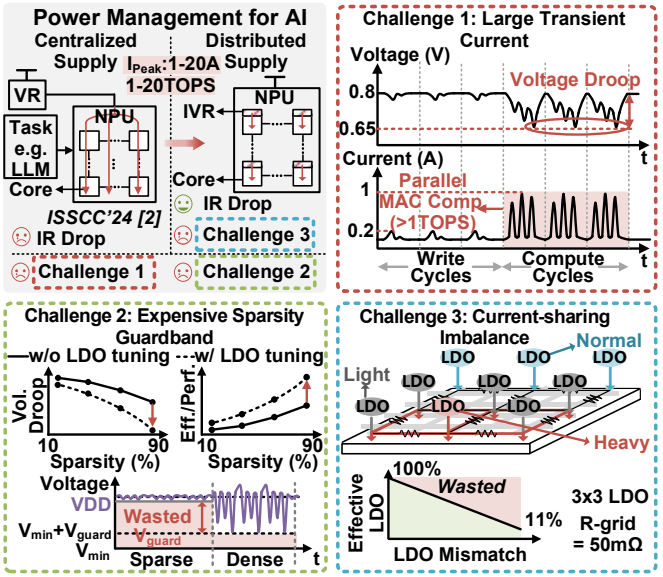


Fig. 1. Distributed power management for AI processors and the key challenges.

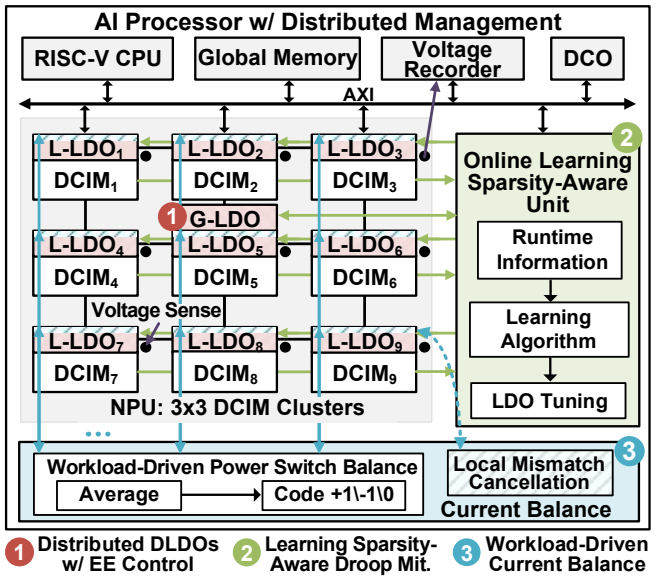


Fig. 2. Overall architecture of AI processor and the distributed power management.

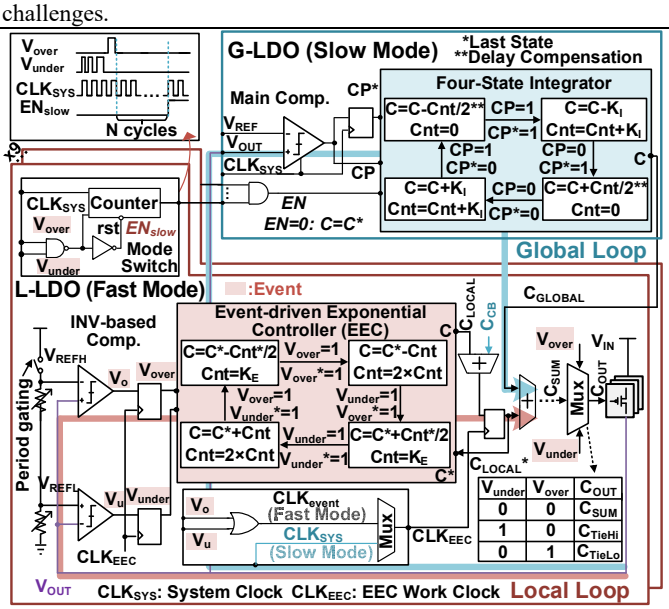


Fig. 3. Dual-loop control with a global loop in the G-LDO for slow mode control and a local loop in the L-LDOs for fast mode control.

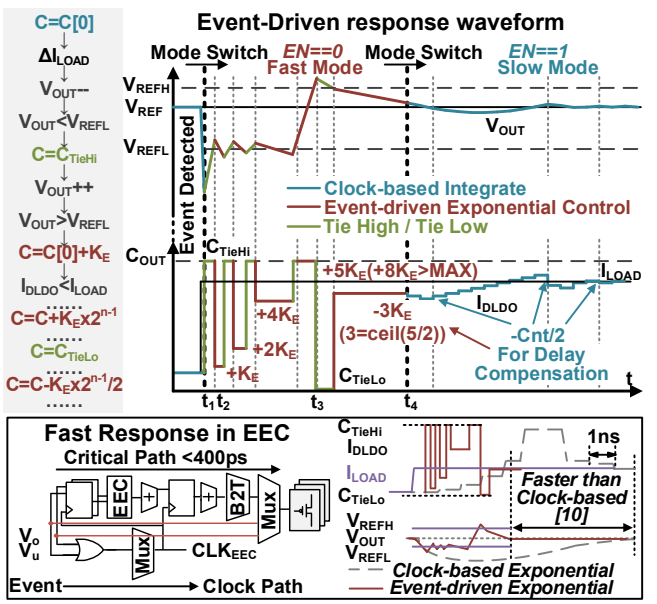


Fig. 4. Event-driven response waveform of our EEC.

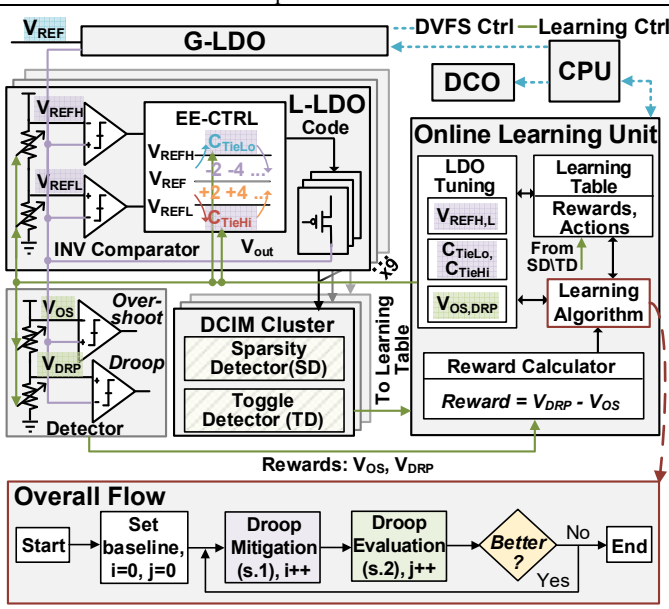


Fig. 5. Online learning-based sparsity-aware droop mitigation.

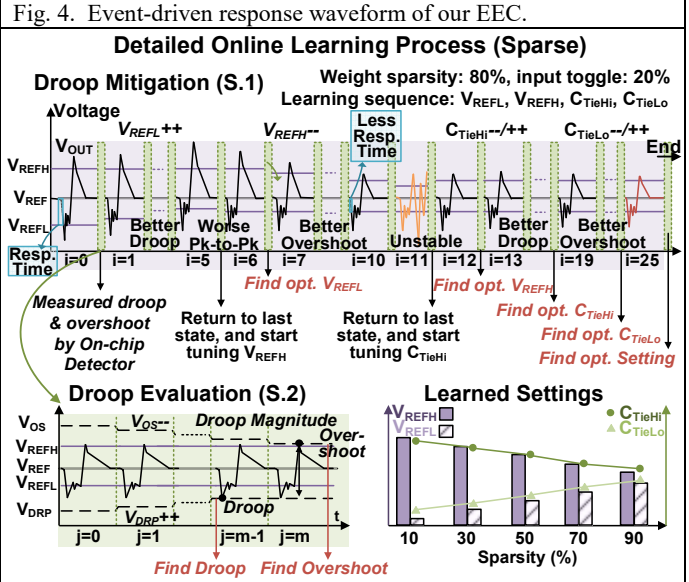


Fig. 6. Learning process for sparsity-aware droop mitigation and learned settings. The sparse workload that has a smaller droop ends up having a narrower detection window between V_{REFH} and V_{REFL}.

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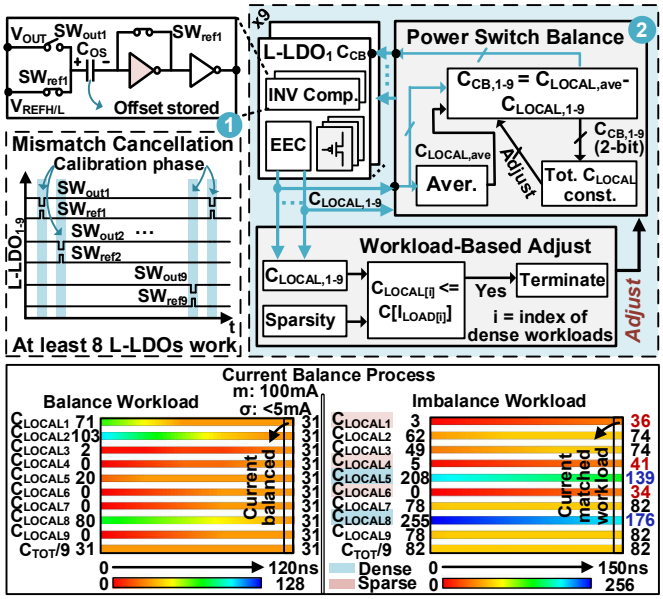


Fig. 7. Workload-driven current balance and mismatch cancellation.

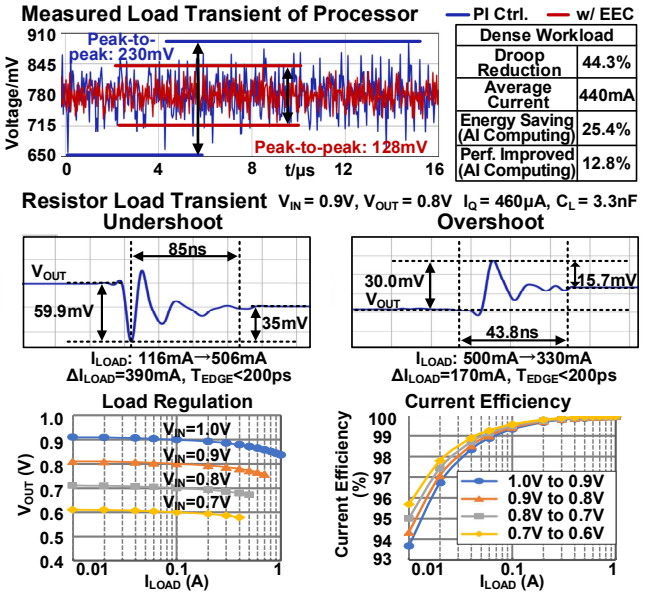


Fig. 8. Measured performance of event-driven exponential LDO.

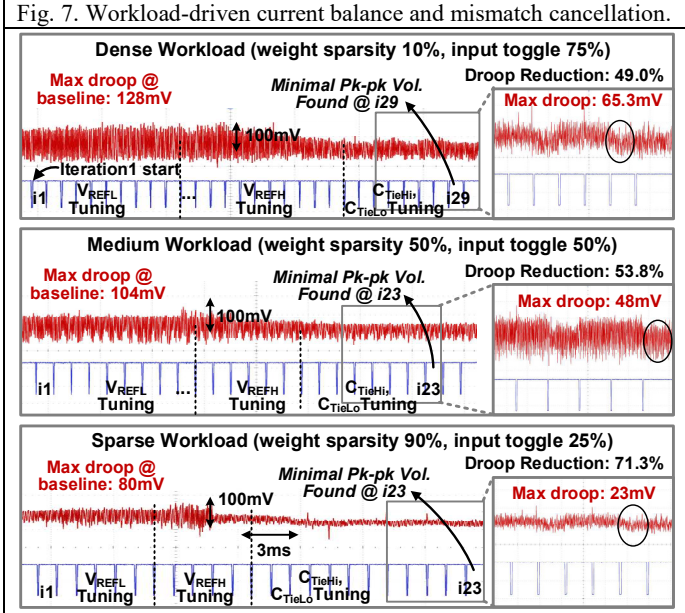


Fig. 9. Measured waveforms for learning-based droop mitigation.

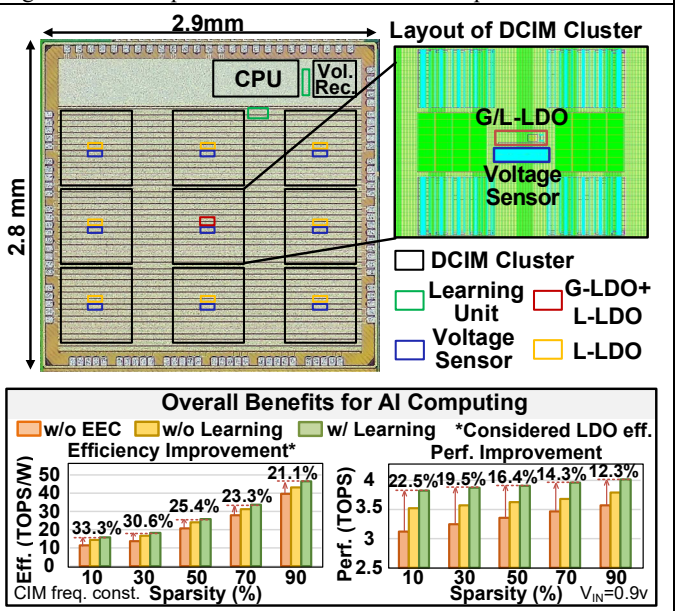


Fig. 10. Die photo and the overall benefits of our scheme.

TABLE I. Comparison table with related designs.

	ISSCC '24 [12]	VLSI '23 [13]	ISSCC '24 [9]	ISSCC '24 [3]	JSSC '21 [4]	This work
Technology	28nm	65nm	12nm	28nm	65nm	**22nm**
Processor	CPU	CPU	AI SoC	N/A	N/A	**AI SoC (DCIM)**
Voltage [V]	0.9	1.8	0.5-1	0.85-1.15	0.5-1	**0.6-1.0**
Frequency	500MHz	400-950MHz	680MHz-1.6GHz	1GHz	1GHz	**70-500MHz**
Management Circuit	Proactive Clock Gating	Buck w/ Machine Learning	DVFS w/ LDO/ TRO	Distribut-ed LDO w/ PI	Distribut-ed LDO w/ ED PI	**Distributed LDO w/ Dual-loop EEC**
Online learning	×	✓	×	×	×	✓
Droop Reduction	32%	50%*1	N/A	N/A	N/A	**71.6% (44.3%: EEC LDO; 27.3%: Learning)**
Performance Improve	-0.6%	9.9%	19-27%	N/A	N/A	**12.3-22.5%*2**
Energy Saving	9.8%*1	9.1%	2.7%*1	N/A	N/A	**21.1-33.3%**
Current balance	N/A	N/A	N/A	✓	×	✓
FOM*3 [ps]	N/A	N/A	N/A	4.94 (T_{EDGE}= 10ns)	3.75 (T_{EDGE}< 200ps)	**0.35 (T_{EDGE}< 200ps)**

1. Estimated based on the reported data
2. Test on an 870M Transformer block from Stable Diffusion v1.5 model
3. FOM = C_{OUT}·($\Delta V_{OUT}/\Delta I_{LOAD}$)·($I_Q/\Delta I_{LOAD}$), smaller FOM is better

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